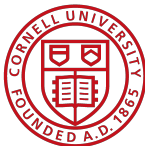


Causal Mediation in Natural Experiments

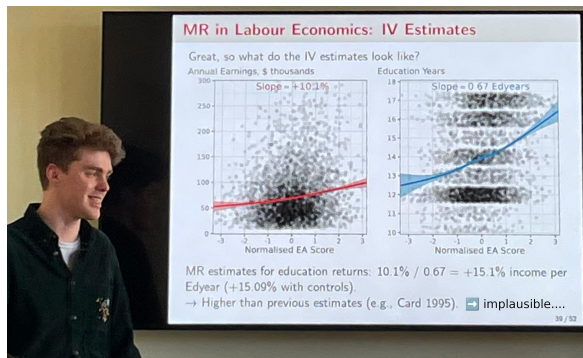
Senan Hogan-Hennessy
Economics Department, Cornell University
seh325@cornell.edu



Cornell Economics, PhD Defence
29 June 2026

About Me

I am an economics PhD student at Cornell University, with research interests in labour economics and applied econometrics.



- Other applied research studying education polygenic index, and mediation of its causal effect through education.

Introduction

Natural experiments are settings with credible estimates of causal effects

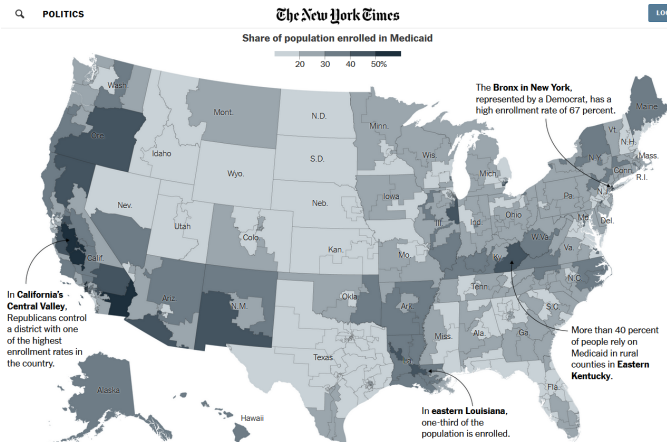
- Little information on the mechanisms through which they operate
- Limits understanding of the decisions and underlying economic system
- Causal Mediation (CM) is a framework for sufficiently analysing a causal effect along an observed mechanism, which is not widely used in applied economics.

This paper:

- 1 Develop selection bias concept for CM in observational settings, with reasoning bias can crowd out inference
- 2 Build an MTE-based model to tackle the identification problem
- 3 Illustrate my methods with decomposing causal effects in the Oregon Health Insurance Experiment.

Oregon Health Insurance Experiment

In the USA, healthcare is only provided by the government in special cases
 → Medicaid is the government programme which provides health insurance for those close to the poverty line (> 70 million people in 2025).



Oregon Health Insurance Experiment

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 → Medicaid is the government programme which provides health insurance
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2004: Oregon Medicaid enrolment frozen



2008: Enrolment reopens...
 90,000 sign up (far exceeding funding allocated)



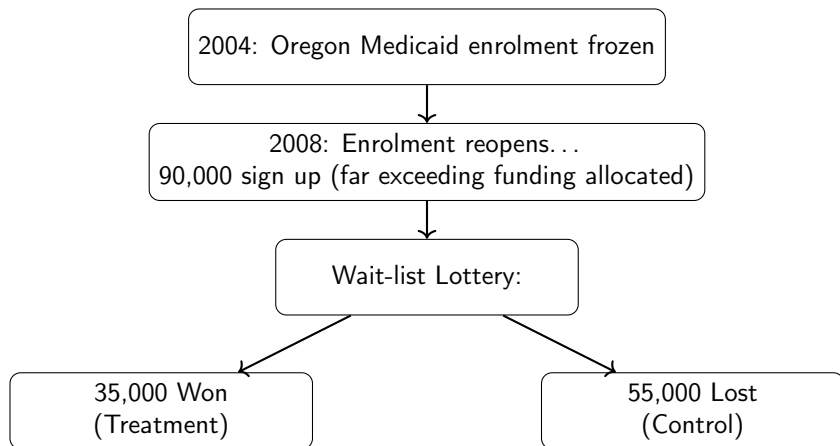
Wait-list Lottery:

35,000 Won
 (Treatment)

55,000 Lost
 (Control)

Oregon Health Insurance Experiment

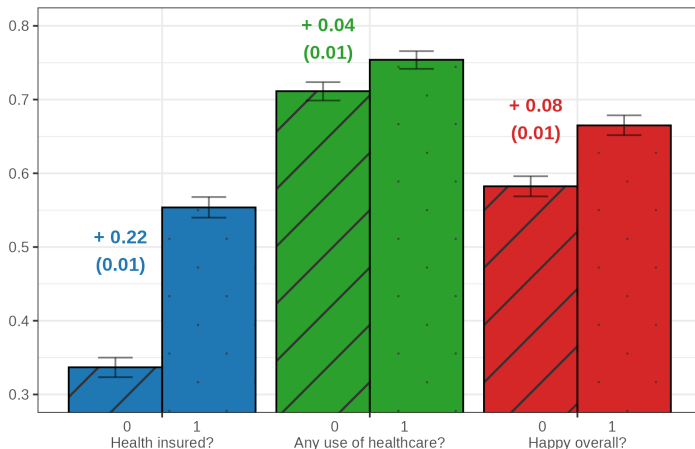
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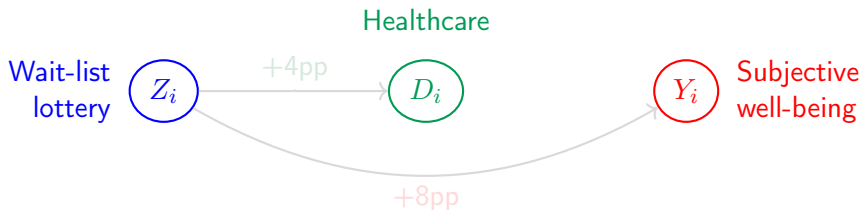
Winning this wait-list lottery significantly increased healthcare usage, plus subjective well-being (Finkelstein et al, 2012).

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Oregon — Suggestive Evidence

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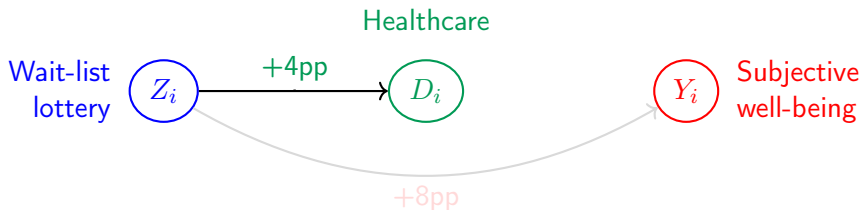


Suggestive evidence:

- If first-stage $\neq 0$, then healthcare may be a mediating mechanism
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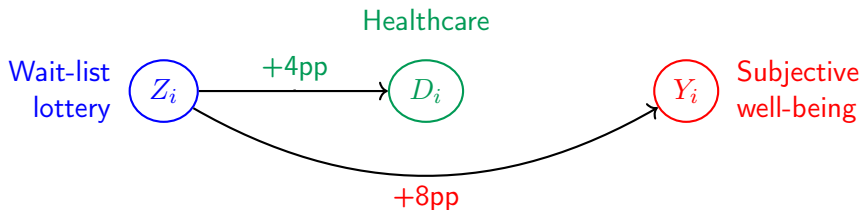


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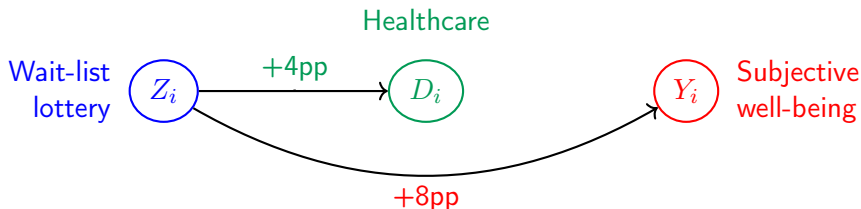


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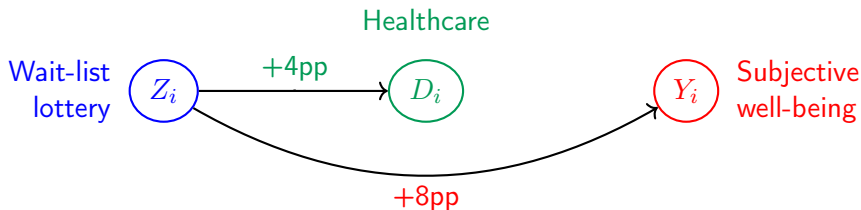


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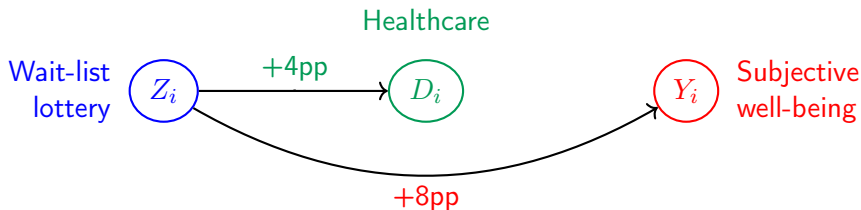


What about direct effects?

- Winning access to Medicaid means you can file for free health insurance (income effect)
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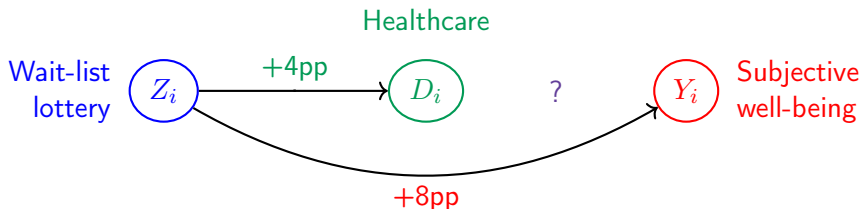


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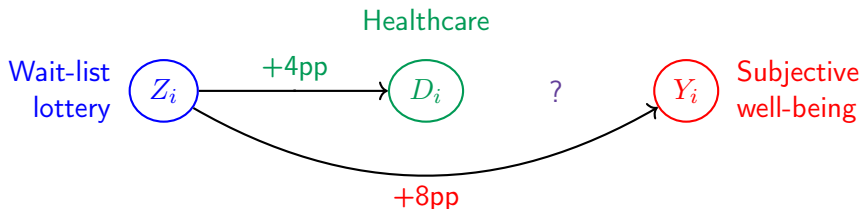
There is one missing piece to make a **definitive conclusion**:

Size of causal effect $D_i \rightarrow Y_i \dots$

- If large, then **healthcare** explains all the lottery effect
- If small/zero then, then all **direct** (e.g., psychological) gains.

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Causal Mediation (CM)

CM is an alternative framework to studying mechanisms, giving sufficient evidence on the mediating mechanism.

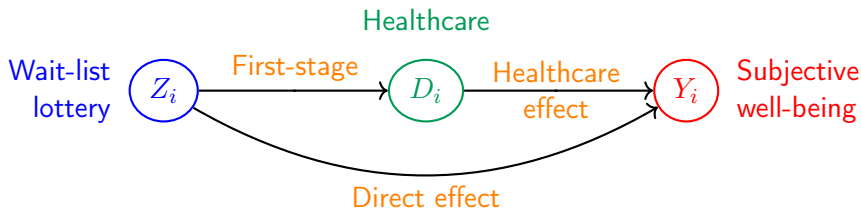


CM aims to decompose the ATE in two channels, direct and indirect effects

$$\text{ATE} = \text{ADE} + \text{AIE}.$$

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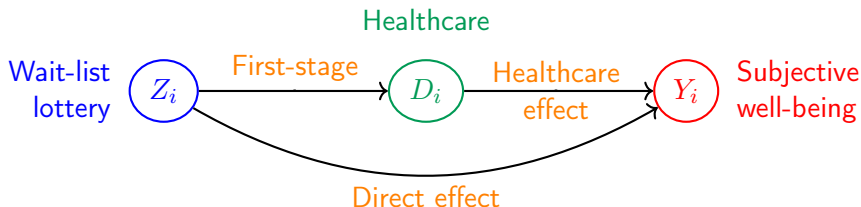


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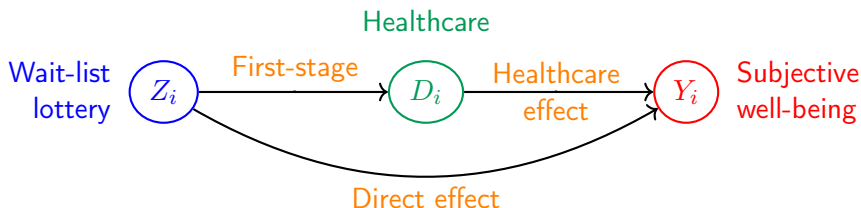
CM decomposes the ATE into

$$\text{Average Indirect Effect (AIE)} : \mathbb{E} \left[Y_i \left(Z_i, D_i(1) \right) - Y_i \left(Z_i, D_i(0) \right) \right]$$

AIE represents the average effect going through healthcare.

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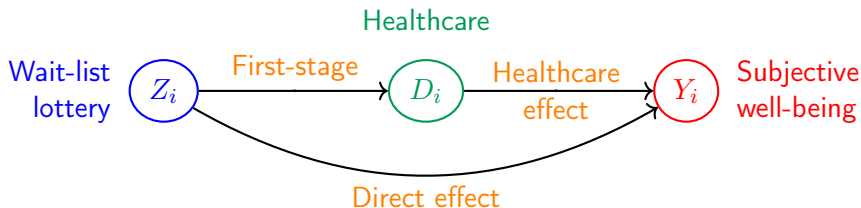
CM decomposes the ATE into

$$\text{Average Direct Effect (ADE)} : \mathbb{E} \left[Y_i \left(1, D_i(Z_i) \right) - Y_i \left(0, D_i(Z_i) \right) \right]$$

ADE represents the average effect going absent healthcare.

Causal Mediation (CM)

Identification of the ADE + AIE needs further assumptions.



Conventional CM relies on two identifying assumptions,

- 1 Treatment Z_i is (quasi-)randomly assigned
- 2 Mediator D_i is (quasi-)randomly assigned, conditional on Z_i realisation and controls X_i .

⇒ ADE + AIE are separately identified by two-stage regression (Imai Keele Yamamoto 2010).

Causal Mediation (CM)

This approach (conventional CM) is used heavily in epidemiology and medicine to give evidence for the channels of a treatment effect, but there is a reason why this is not prominent in economics.

Identifying assumptions:

- ① **Treatment** Z_i is (quasi-)randomly assigned
- ② **Mediator** D_i is (quasi-)randomly assigned, conditional on Z_i realisation (and covariates $\mathbf{X}_i$).

Translation: Healthcare is a quasi-random choice, conditional on wait-list lottery realisation and demographic controls.

Would this be plausible in settings economists study?

Causal Mediation (CM) — Roy Model

Consider the case that people, after the lottery, choose to **visit the doctor in the next 12 months** based on subjective costs and benefits,

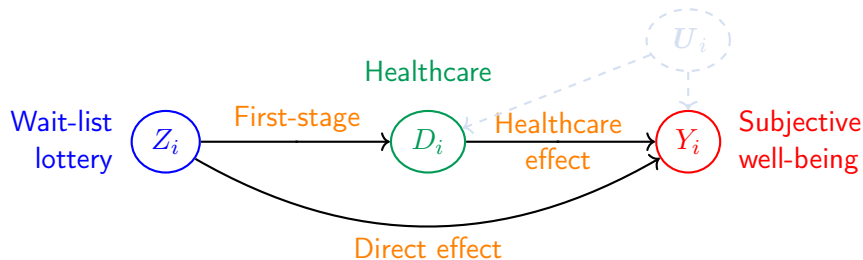
$$D_i(z') = \mathbb{1} \left\{ \underbrace{C_i}_{\text{Costs}} \leq \underbrace{Y_i(z', 1) - Y_i(z', 0)}_{\text{Benefits}} \right\}.$$

The **wait-list lottery** has no strategic selection, but then **visiting healthcare** is a choice.

Theorem: If choice to attend healthcare is based on costs and benefits (Roy model) and control variables do not explain all benefits $\implies$ **mediator mechanism** is not random, there is unobserved confounding.

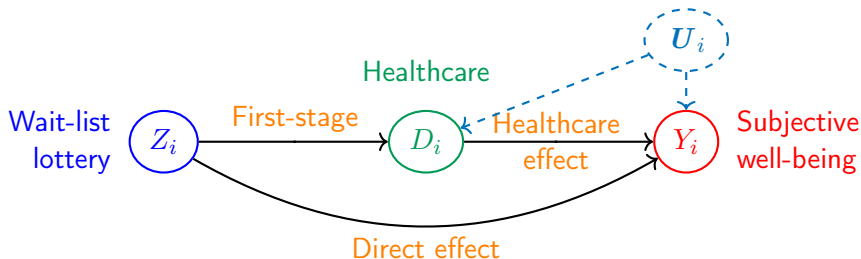
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Individual unobserved benefits are an unobserved confounder U_i ; here,



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In economic settings, Conventional CM analyses have bias similar to classical selection bias (Heckman Ichimura Smith Todd 1998).

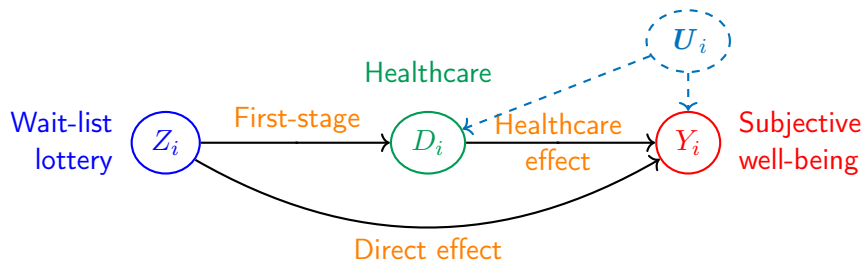
- Direct: CM Estimand = ADE + (Selection Bias + Group difference bias)
- Indirect: CM Estimand = AIE + (Selection Bias + Group difference bias)

► ADE biases

► AIE biases

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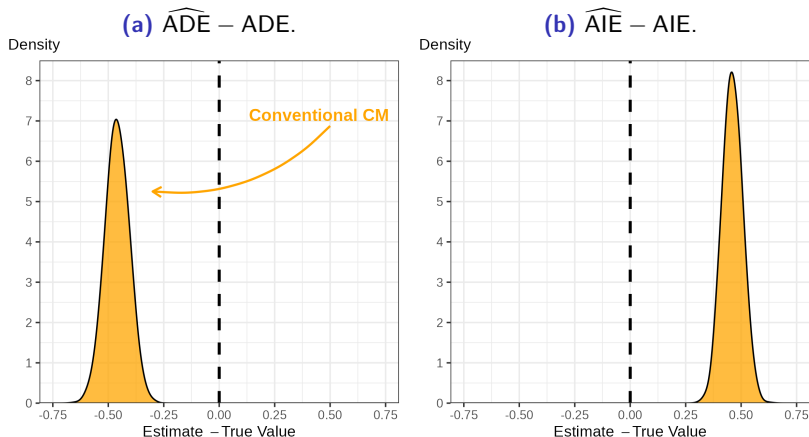
► ADE biases

► AIE biases

Causal Mediation (CM) — Selection Bias

With strategic selection, the bias terms can be large and mislead inference on how much goes through the mediating channel.

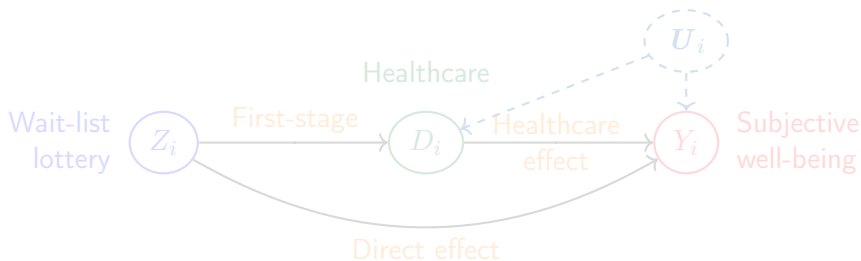
Figure: Simulated Distribution from 10,000 simulated datasets.



CM with Selection

Conventional CM does not identify ADE + AIE in economic settings, so I build a structural model for natural experiment settings.

Take as given that Z_i is quasi-randomly assigned, but D_i is not:

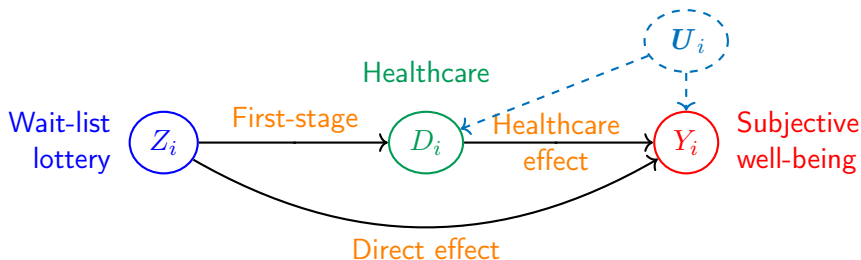


Intuition: model U_i via healthcare MTE to identify ADE + AIE.

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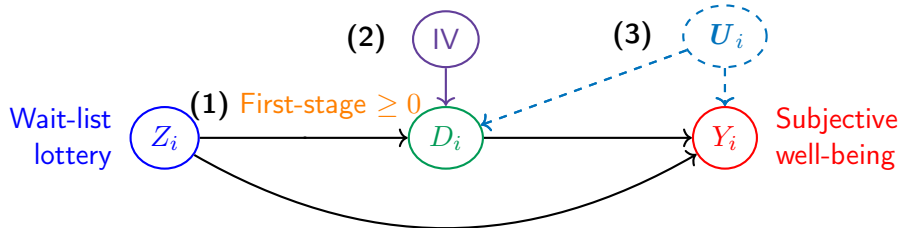
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MTE Model

The structural model is based on 3 assumptions.

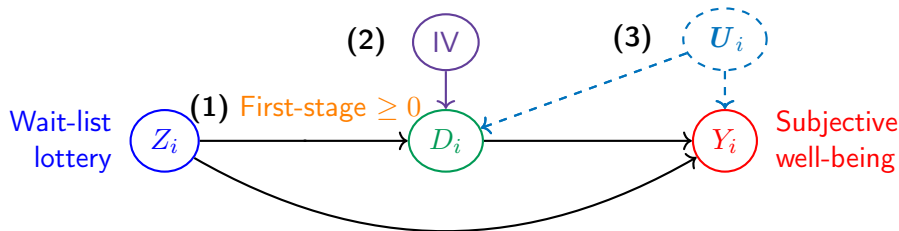


(1) First-stage monotonicity,

Intuition: No defiers — no one visits healthcare less if winning wait-list lottery, relative to losing.

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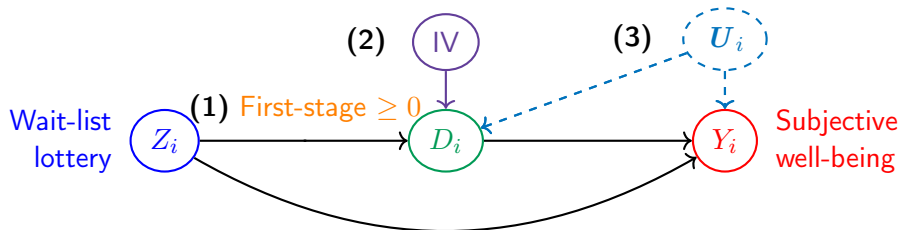


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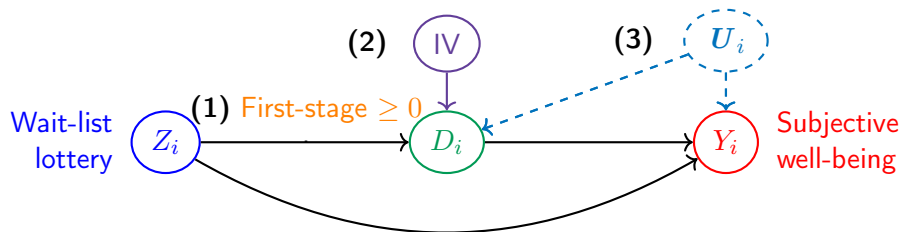
(2) Mediator take-up cost IV

Requires an IV , which affects Y_i only via D_i .

Key example: Cost-shifting IV — random variation in **healthcare take-up** (not gains), e.g. **different healthcare costs**.

MTE Model

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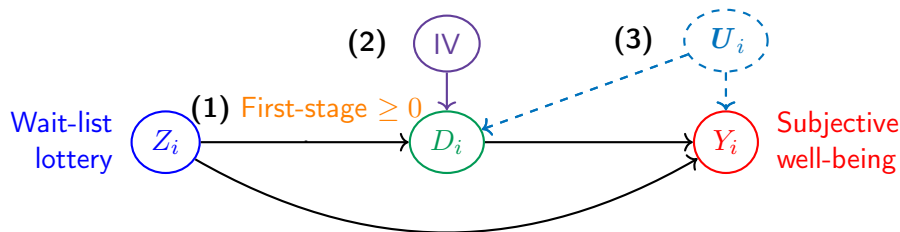


(3) Selection on benefits — unobserved selection in D_i is relevant

Key example: Roy model, people choose to take **healthcare** if internal **subjective gains** exceed costs.

MTE Model — Identification

The structural model is based on 3 assumptions.



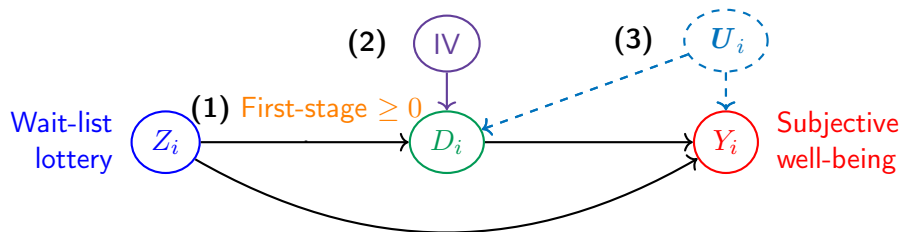
Theorem: Under assumptions (1), (2), (3) ADE + AIE are identified.

ADE Intuition: Control for unobserved confounding via healthcare MTE.

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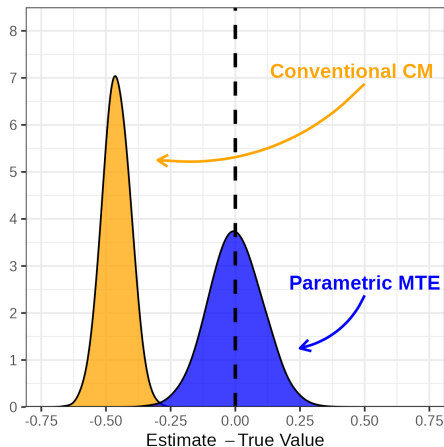
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MTE Model — Estimation

Figure: CM Estimates from 10,000 simulated datasets with **Normal** Errors.

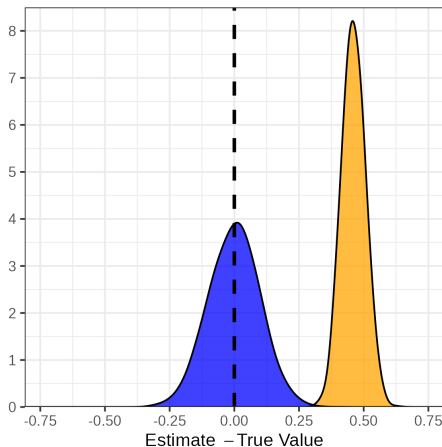
(a) $\widehat{ADE} - ADE$.

Density



(b) $\widehat{AIE} - AIE$.

Density

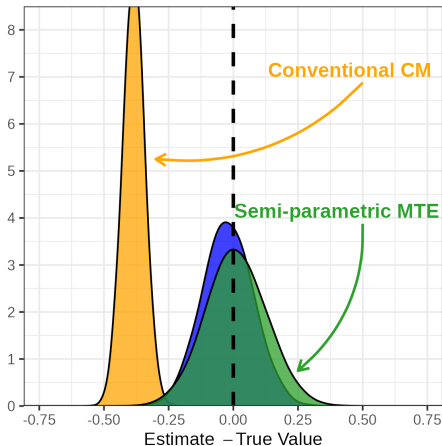


MTE Model — Estimation

Figure: CM Estimates from 10,000 simulated datasets with **Uniform** Errors.

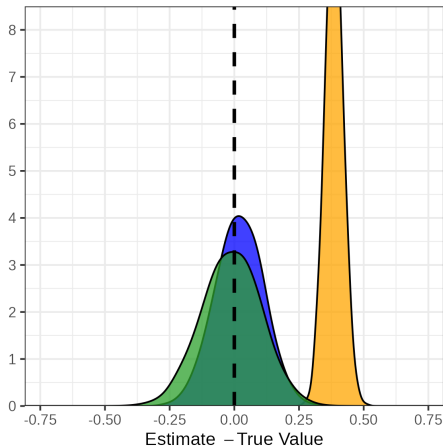
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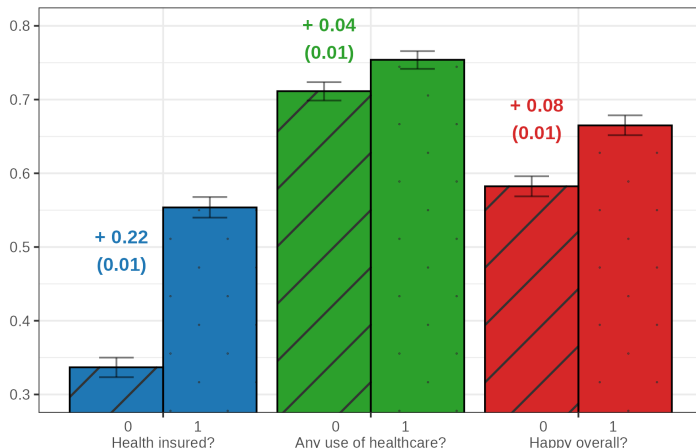
Density



Return to Oregon

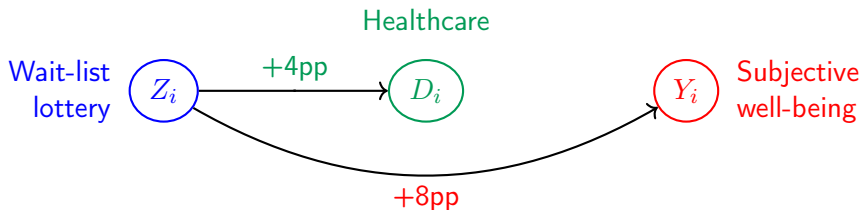
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Mean Outcome, winning or losing the wait-list lottery.



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Suggestive evidence:

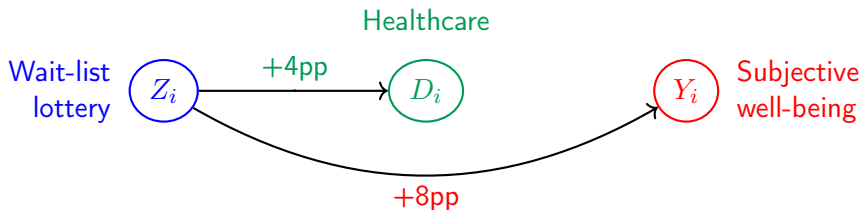
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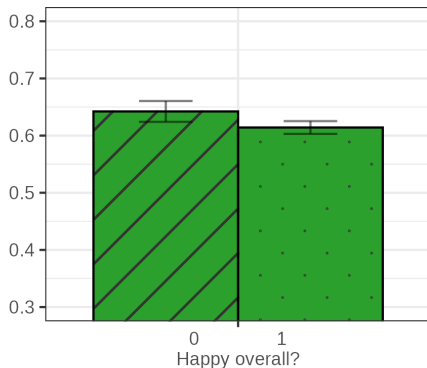
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Oregon — Conventional CM

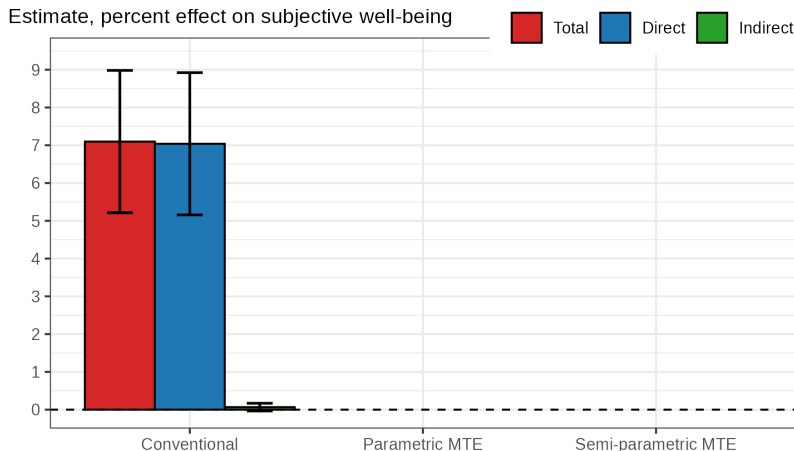
Does using healthcare improve subjective well-being?



- OLS estimate of healthcare $D_i \rightarrow Y_i$ subjective well-being is -2.5pp (1.1).
- Controls for initial health conditions gives $+2.8\text{pp}$ (1.1).

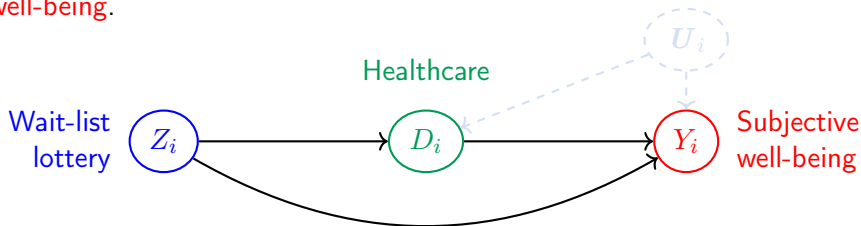
Oregon — Conventional CM

Conventional CM estimates lottery **subjective well-being** effects as mostly direct, ≈ 0 **healthcare**.



Oregon — Selection Bias

OLS Estimates had negative or little effects of **healthcare** → **subjective well-being**.

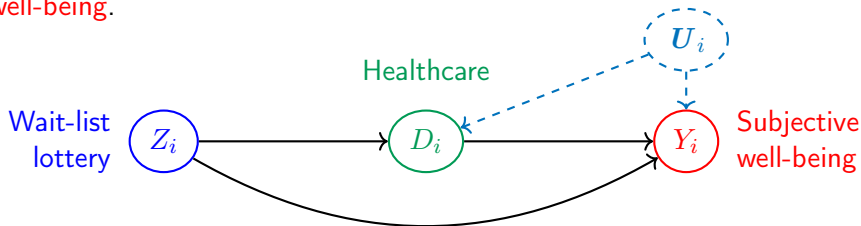


Example confounder: undiagnosed conditions among those without health insurance, near or below the poverty line.

Implication: negative selection bias in OLS estimates, and Conventional CM underestimated **indirect healthcare channel**.

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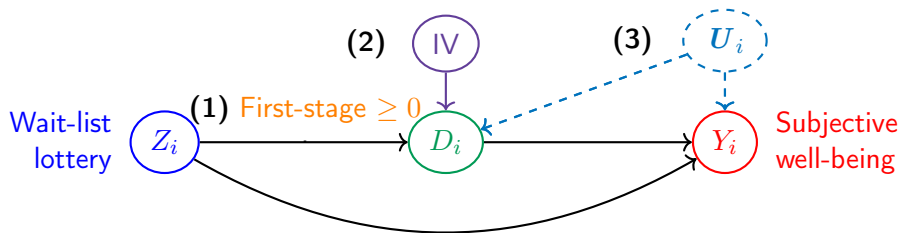


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Oregon — MTE Model

I bring the MTE model to these data instead.



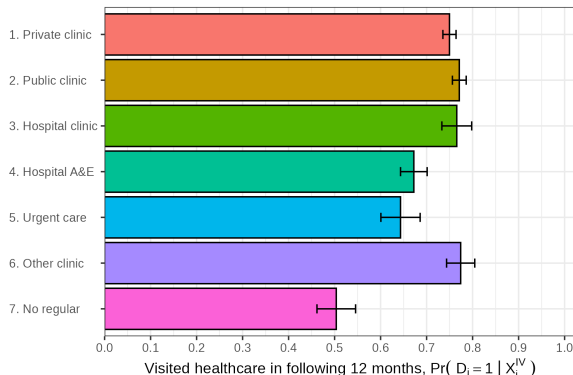
Healthcare IV: pre-lottery healthcare location.

Intuition: Different locations charge different prices for similar healthcare. Heading to A&E costs more than a local doctor's office.

Oregon — MTE Model

Oregon Health Insurance applicants asked pre-lottery healthcare location.

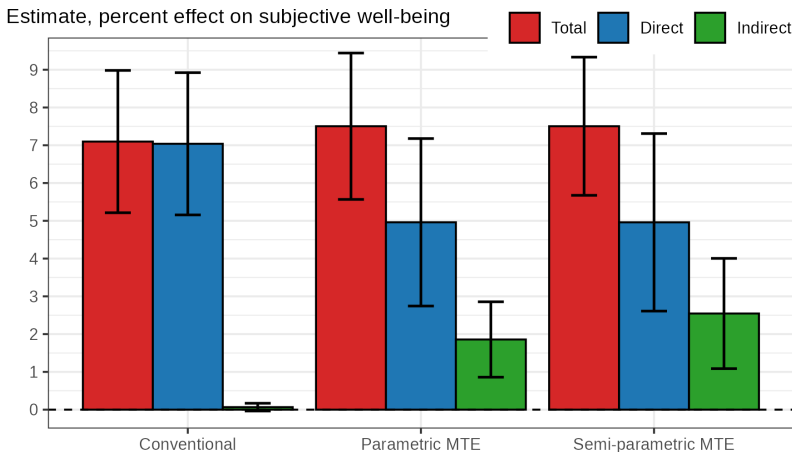
Usual Healthcare Location



IV assumption: where the uninsured, near poverty line, participants visits is indicative of their local healthcare access and cost.

Oregon — MTE Model

Using MTE approach, with regular healthcare location IV, restores indirect effect through increasing healthcare visitation.



Conclusion

Overview:

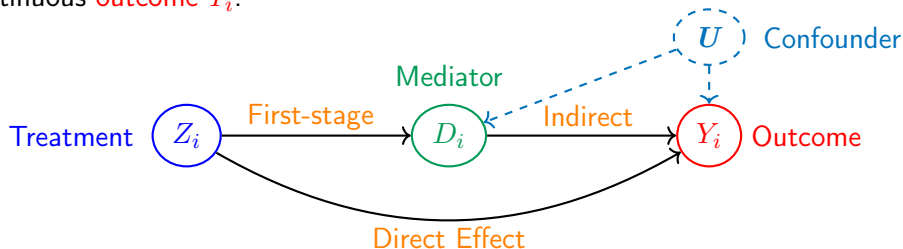
- ① Selection bias in conventional CM analyses when mediator mechanism strategically selected into.
- ② Connect CM with labour theory + selection-into-treatment + MTEs.

Caveats and points to remember:

- Structural assumptions and IV for identification + estimation (not ideal)
- Application to Oregon Health Insurance Experiment, showing **subjective well-being** effects mediated by **healthcare take-up**
- **Credible** analyses of mechanisms are hard in practice, and wide confidence intervals show true uncertainty.

Appendix: CM Guiding Model

Consider binary **treatment** $Z_i = 0, 1$, binary **mediator** $D_i = 0, 1$, and continuous **outcome** Y_i .



Average Direct Effect (ADE): $\mathbb{E} \left[Y_i \left(\mathbf{1}, D_i(Z_i) \right) - Y_i \left(\mathbf{0}, D_i(Z_i) \right) \right]$

- ADE is causal effect $Z \rightarrow Y$, blocking the indirect D_i path.

Average Indirect Effect (AIE): $\mathbb{E} \left[Y_i \left(Z_i, \mathbf{D}_i(1) \right) - Y_i \left(Z_i, \mathbf{D}_i(0) \right) \right]$

- AIE is causal effect of $D_i(Z_i) \rightarrow Y_i$, blocking the direct Z_i path.

Group Difference — ADE

CM effects contaminated by (less interpretable) bias terms.

$$\text{CM Estimand} = \text{ADEM} + \text{Selection Bias}$$

$$\begin{aligned}
 & \underbrace{\mathbb{E}_{D_i} \left[\mathbb{E} [Y_i \mid Z_i = 1, D_i] - \mathbb{E} [Y_i \mid Z_i = 0, D_i] \right]}_{\text{Estimand, Direct Effect}} \\
 &= \underbrace{\mathbb{E}_{D_i=d'} \left[\mathbb{E} [Y_i(1, D_i(Z_i)) - Y_i(0, D_i(Z_i)) \mid D_i(1) = d'] \right]}_{\text{Average Direct Effect on Mediator (ADEM) take-up — i.e., } D_i(1) \text{ weighted}} \\
 & \quad + \underbrace{\mathbb{E}_{D_i} \left[\mathbb{E} [Y_i(0, D_i(Z_i)) \mid D_i(1) = d'] - \mathbb{E} [Y_i(0, D_i(Z_i)) \mid D_i(0) = d'] \right]}_{\text{Selection Bias}}
 \end{aligned}$$

The weighted ADE you get here is a positive weighted sum of local ADEs, but with policy irrelevant weights $D_i(1) = d'$.

⇒ consider this group bias, noting difference from true ADE. [▶ Back](#)

Selection Bias — Direct Effect

CM Effects + contaminating bias.

$$\text{CM Estimand} = \text{ADE} + \left(\text{Selection Bias} + \text{Group difference bias} \right)$$

► Model

$$\begin{aligned} & \underbrace{\mathbb{E}_{D_i=d'} \left[\mathbb{E} [Y_i \mid Z_i = 1, D_i = d'] - \mathbb{E} [Y_i \mid Z_i = 0, D_i = d'] \right]}_{\text{Estimand, Direct Effect}} \\ &= \underbrace{\mathbb{E} [Y_i(1, D_i(Z_i)) - Y_i(0, D_i(Z_i))]}_{\text{Average Direct Effect}} \\ &+ \underbrace{\mathbb{E}_{D_i=d'} \left[\mathbb{E} [Y_i(0, D_i(Z_i)) \mid D_i(1) = d'] - \mathbb{E} [Y_i(0, D_i(Z_i)) \mid D_i(0) = d'] \right]}_{\text{Selection Bias}} \\ &+ \underbrace{\mathbb{E}_{D_i=d'} \left[\left(1 - \Pr(D_i(1) = d') \right) \right.}_{\text{Group difference bias}} \\ &\quad \times \left. \left(\mathbb{E} [Y_i(1, D_i(Z_i)) - Y_i(0, D_i(Z_i)) \mid D_i(1) = 1 - d'] \right. \right. \\ &\quad \left. \left. - \mathbb{E} [Y_i(1, D_i(Z_i)) - Y_i(0, D_i(Z_i)) \mid D_i(0) = d'] \right) \right]}_{\text{Group-diff}} \end{aligned}$$

Group Difference — AIE

CM effects contaminated by (less interpretable) bias terms.

$$\text{CM Estimand} = \text{AIEM} + \left(\text{Selection Bias} + \text{Group difference bias} \right)$$

$$\begin{aligned} & \underbrace{\mathbb{E}_{Z_i} \left[\left(\mathbb{E}[D_i | Z_i = 1] - \mathbb{E}[D_i | Z_i = 0] \right) \times \left(\mathbb{E}[Y_i | Z_i, D_i = 1] - \mathbb{E}[Y_i | Z_i, D_i = 0] \right) \right]}_{\text{Estimand, Indirect Effect}} \\ &= \underbrace{\mathbb{E} \left[Y_i(Z_i, D_i(1)) - Y_i(Z_i, D_i(0)) \mid D_i = 1 \right]}_{\text{Average Indirect Effect on Mediated (AIEM) — i.e., } D_i = 1 \text{ weighted}} \\ &+ \underbrace{\pi \left(\mathbb{E}[Y_i(Z_i, 0) \mid D_i = 1] - \mathbb{E}[Y_i(Z_i, 0) \mid D_i = 0] \right)}_{\text{Selection Bias}} \\ &+ \underbrace{\pi \left[\left(\frac{1 - \Pr(D_i(1) = 1, D_i(0) = 0)}{\Pr(D_i(1) = 1, D_i(0) = 0)} \right) \left(\mathbb{E}[Y_i(Z_i, 1) - Y_i(Z_i, 0) \mid D_i(1) = 0 \text{ or } D_i(0) = 1] - \mathbb{E}[Y_i(Z_i, 1) - Y_i(Z_i, 0)] \right) \right]}_{\text{Groups difference Bias}} \end{aligned}$$

The weighted AIE you get here is not a positive weighted sum of local AIEs, because the AIE is only about $D(Z)$ compliers. [► Model](#).

⇒ consider this group bias, noting difference from true AIE.

Selection Bias — Indirect Effect

CM Effects + contaminating bias, where $\bar{\pi} = \Pr(D_i(0) \neq D_i(1))$.

$$\text{CM Estimand} = \text{AIE} + \left(\text{Selection Bias} + \text{Group difference bias} \right) \quad \text{► Model}$$

$$\begin{aligned} & \underbrace{\mathbb{E}_{Z_i} \left[\left(\mathbb{E}[D_i | Z_i = 1] - \mathbb{E}[D_i | Z_i = 0] \right) \times \left(\mathbb{E}[Y_i | Z_i, D_i = 1] - \mathbb{E}[Y_i | Z_i, D_i = 0] \right) \right]}_{\text{Estimand, Indirect Effect}} \\ &= \underbrace{\mathbb{E}[Y_i(Z_i, D_i(1)) - Y_i(Z_i, D_i(0))]}_{\text{Average Indirect Effect}} \\ &+ \underbrace{\bar{\pi} \left(\mathbb{E}[Y_i(Z_i, 0) | D_i = 1] - \mathbb{E}[Y_i(Z_i, 0) | D_i = 0] \right)}_{\text{Selection Bias}} \\ &+ \bar{\pi} \left[\begin{aligned} & \left(1 - \Pr(D_i = 1) \right) \left(\mathbb{E}[Y_i(Z_i, 1) - Y_i(Z_i, 0) | D_i = 1] \right. \\ & \quad \left. - \mathbb{E}[Y_i(Z_i, 1) - Y_i(Z_i, 0) | D_i = 0] \right) \\ & + \left(\frac{1 - \Pr(D_i(1) = 1, D_i(0) = 0)}{\Pr(D_i(1) = 1, D_i(0) = 0)} \right) \left(\mathbb{E}[Y_i(Z_i, 1) - Y_i(Z_i, 0) | D_i(Z_i) \neq Z_i] \right. \\ & \quad \left. - \mathbb{E}[Y_i(Z_i, 1) - Y_i(Z_i, 0)] \right) \end{aligned} \right] \end{aligned}$$

Groups difference Bias ► Group-diff

Semi-parametric Control Functions

Semi-parametric specifications for the CFs λ_0, λ_1 bring some complications to estimating the AIE.

$$\mathbb{E}[Y_i | Z_i, D_i = 0, \mathbf{X}_i] = \alpha + \gamma Z_i + \varphi(\mathbf{X}_i) + \rho_0 \lambda_0(\pi(Z_i; \mathbf{X}_i)),$$

$$\mathbb{E}[Y_i | Z_i, D_i = 1, \mathbf{X}_i] = (\alpha + \beta) + (\gamma + \delta) Z_i + \varphi(\mathbf{X}_i) + \rho_1 \lambda_1(\pi(Z_i; \mathbf{X}_i))$$

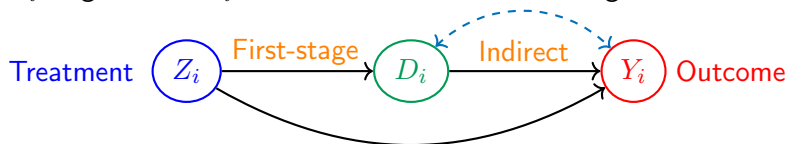
Intercepts, $\alpha, (\alpha + \beta)$, and relevance parameters ρ_0, ρ_1 are not separately identified from the CFs $\lambda_0(\cdot), \lambda_1(\cdot)$ so CF extrapolation term $(\rho_1 - \rho_0)\Gamma(\pi(0; \mathbf{X}_i), \pi(1; \mathbf{X}_i))$ is not directly identified or estimable.

These problems can be avoided by estimating the AIE using its relation to the ATE, $\widehat{\text{AIE}}^{\text{CF}} =$

$$\widehat{\text{ATE}} - (1 - \bar{Z}) \underbrace{\left(\frac{1}{N} \sum_{i=1}^N \hat{\gamma} + \hat{\delta} \hat{\pi}(1; \mathbf{X}_i) \right)}_{\widehat{\text{ADE}} \text{ given } Z_i=1} - \bar{Z} \underbrace{\left(\frac{1}{N} \sum_{i=1}^N \hat{\gamma} + \hat{\delta} \hat{\pi}(0; \mathbf{X}_i) \right)}_{\widehat{\text{ADE}} \text{ given } Z_i=0}.$$

Appendix: CM with Selection

Suppose Z_i is ignorable, D_i is not, so we have the following causal model.



Then this system has the following random coefficient equations:

$$D_i = \phi + \pi Z_i + \varphi(\mathbf{X}_i) + U_i$$

$$Y_i = \alpha + \beta D_i + \gamma Z_i + \delta Z_i D_i + \zeta(\mathbf{X}_i) + \underbrace{(1 - D_i) U_{0,i} + D_i U_{1,i}}_{\text{Correlated error term}}$$

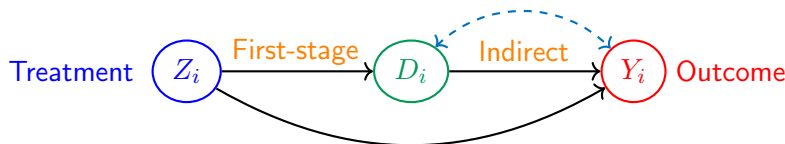
where β, γ, δ are functions of $\mu_{d'}(z'; \mathbf{X}_i)$.

$$\text{ADE} = \mathbb{E}[\gamma + \delta D_i], \quad \text{AIE} = \mathbb{E}\left[\pi(\beta + \delta Z_i + \tilde{U}_i)\right]$$

with $\tilde{U}_i = \mathbb{E}[U_{1,i} - U_{0,i} | \mathbf{X}_i, D_i(0) \neq D_i(1)]$ unobserved complier gains.

Appendix: CM with Selection

Suppose Z_i is ignorable, D_i is not, so we have the following causal model.



Main problem, second-stage is not identified:

$$\begin{aligned} \mathbb{E}[Y_i | Z_i, D_i, \mathbf{X}_i] &= \alpha + \beta D_i + \gamma Z_i + \delta Z_i D_i + \varphi(\mathbf{X}_i) \\ &+ (1 - D_i) \mathbb{E}[U_{0,i} | D_i = 0, \mathbf{X}_i] \\ &+ \underbrace{D_i \mathbb{E}[U_{1,i} | D_i = 1, \mathbf{X}_i]}_{\text{Unobserved } D_i \text{ confounding.}} \end{aligned}$$

Identification intuition: Identify second-stage via MTE control function.

Appendix: CM with Selection — Identification

Assume:

- ① Mediator monotonicity, $\Pr(D_i(0) \leq D_i(1) \mid \mathbf{X}_i) = 1$
 $\implies D_i(z') = \mathbb{1}\{U_i \leq \pi(z'; \mathbf{X}_i)\}, \text{ for } z' = 0, 1 \text{ (Vycatil 2002).}$
- ② Selection on mediator benefits, $\text{Cov}(U_i, U_{0,i}), \text{Cov}(U_i, U_{1,i}) \neq 0$
 $\implies$ First-stage take-up informs second-stage confounding.
- ③ There is an IV for the mediator, $\mathbf{X}_i^{\text{IV}}$ among control variables $\mathbf{X}_i$.
 $\implies \pi(Z_i; \mathbf{X}_i) = \Pr(D_i = 1 \mid Z_i, \mathbf{X}_i)$ is separately identified.

Proposition:

$$\begin{aligned} & \mathbb{E}[Y_i(z', 1) - Y_i(z', 0) \mid Z_i = z', \mathbf{X}_i, U_i = p'] \\ &= \beta + \delta z' + \mathbb{E}[U_{1,i} - U_{0,i} \mid \mathbf{X}_i, U_i = p'], \quad \text{for } p' \in (0, 1). \end{aligned}$$

Appendix: CM with Selection — Identification

The marginal effect has corresponding Control Functions (CFs), describing unobserved selection-into- D_i ,

$$\rho_0 \lambda_0(p') = \mathbb{E} [U_{0,i} \mid p' \leq U_i], \quad \rho_1 \lambda_1(p') = \mathbb{E} [U_{1,i} \mid U_i \leq p'] .$$

These CFs restore second-stage identification, by extrapolating from $\mathbf{X}_i^{IV}$ compliers to $D_i(Z_i)$ mediator compliers,

$$\begin{aligned} \mathbb{E} [Y_i \mid Z_i, D_i, \mathbf{X}_i] &= \alpha + \beta D_i + \gamma Z_i + \delta Z_i D_i + \varphi(\mathbf{X}_i) \\ &\quad + \underbrace{\rho_0 (1 - D_i) \lambda_0(\pi(Z_i; \mathbf{X}_i)) + \rho_1 D_i \lambda_1(\pi(Z_i; \mathbf{X}_i))}_{\text{CF adjustment.}} \end{aligned}$$

This adjusted second-stage re-identifies the ADE and AIE,

$$\text{ADE} = \mathbb{E} [\gamma + \delta D_i], \quad \text{AIE} = \mathbb{E} \left[\bar{\pi} \left(\beta + \delta Z_i + \underbrace{(\rho_1 - \rho_0) \Gamma(\pi(0; \mathbf{X}_i), \pi(1; \mathbf{X}_i))}_{\text{Mediator compliers extrapolation.}} \right) \right]$$

Appendix: CM with Selection — Estimation

Will explain how estimation works, with simulation evidence.

- ① Random treatment $Z_i \sim \text{Binom}(0.5)$, for $n = 5,000$.
- ② $(U_{0,i}, U_{1,i}) \sim \text{BivariateNormal}(0, 0, \sigma_0, \sigma_1, \rho)$, Costs $C_i \sim N(0, 0.5)$.

Roy **selection-into- D_i** , with constant partial effects + interaction term.

$$D_i(z') = \mathbb{1} \{C_i \leq Y_i(z', 1) - Y_i(z', 0)\},$$

$$Y_i(z', d') = (z' + d' + z'd') + U_{d'} \quad \text{for } z', d' = 0, 1.$$

Following the previous, these data have the following first and second-stage equations, where $\mathbf{X}_i^{\text{IV}}$ is an additive cost IV:

$$D_i = \mathbb{1} \left\{ C_i - (U_{1,i} - U_{0,i}) \leq Z_i - \mathbf{X}_i^{\text{IV}} \right\}$$

$$Y_i = Z_i + D_i + Z_i D_i + (1 - D_i) U_{0,i} + D_i U_{1,i}.$$

$\implies$ unobserved confounding by BivariateNormal $(U_{0,i}, U_{1,i})$.

Appendix: CM with Selection — Estimation

Errors are normal, so system is Heckman (1979) selection model.

CFs are the inverse Mills ratio, with $\phi(\cdot)$ normal pdf and $\Phi(\cdot)$ normal cdf,

$$\lambda_0(p') = \frac{\phi(-\Phi^{-1}(p'))}{\Phi(-\Phi^{-1}(p'))}, \quad \lambda_1(p') = \frac{\phi(\Phi^{-1}(p'))}{\Phi(\Phi^{-1}(p'))}, \quad \text{for } p' \in (0, 1).$$

Parametric Estimation Recipe:

- ① Estimate first-stage $\pi(Z_i; \mathbf{X}_i)$ with probit, including $\mathbf{X}_i^{\text{IV}}$.
- ② Include λ_0, λ_1 CFs in second-stage OLS estimation.
- ③ Compose CM estimates from two-stage plug-in estimates.

→ Same as conventional CM estimates (two-stages), with CFs added.

$$\widehat{\text{ADE}} = \mathbb{E} \left[\widehat{\gamma} + \widehat{\delta} D_i \right], \quad \widehat{\text{AIE}} = \mathbb{E} \left[\widehat{\pi} \left(\widehat{\beta} + \widehat{\delta} Z_i + \underbrace{(\widehat{\rho}_1 - \widehat{\rho}_0) \Gamma(\widehat{\pi}(0; \mathbf{X}_i), \widehat{\pi}(1; \mathbf{X}_i))}_{\text{Mediator compliers extrapolation.}} \right) \right]$$

Appendix: CM with Selection — Estimation

If errors are not normal, then CFs do not have a known form, so semi-parametrically estimate them (e.g., splines).

$$\mathbb{E}[Y_i | Z_i, D_i = 0, \mathbf{X}_i] = \alpha + \gamma Z_i + \varphi(\mathbf{X}_i) + \rho_0 \lambda_0(\pi(Z_i; \mathbf{X}_i)),$$

$$\mathbb{E}[Y_i | Z_i, D_i = 1, \mathbf{X}_i] = (\alpha + \beta) + (\gamma + \delta) Z_i + \varphi(\mathbf{X}_i) + \rho_1 \lambda_1(\pi(Z_i; \mathbf{X}_i))$$

Semi-parametric Estimation Recipe:

- 1 Estimate first-stage $\pi(Z_i; \mathbf{X}_i)$, including $\mathbf{X}_i^{IV}$.
- 2 Estimate second-stage separately for $D_i = 0$ and $D_i = 1$, with regressors $\lambda_0(p')$, $\lambda_1(p')$, semi-parametric in $\hat{\pi}(Z_i; \mathbf{X}_i)$.
- 3 Compose CM estimates from two-stage plug-in estimates.

→ Same as conventional CM estimates, with semi-parametric CFs. ▶ CFs.

$$\widehat{\text{ADE}} = \mathbb{E}[\hat{\gamma} + \hat{\delta} D_i], \quad \widehat{\text{AIE}} = \mathbb{E}\left[\hat{\pi}\left(\hat{\beta} + \hat{\delta} Z_i + (\hat{\rho}_1 - \hat{\rho}_0) \Gamma(\hat{\pi}(0; \mathbf{X}_i), \hat{\pi}(1; \mathbf{X}_i))\right)\right]$$

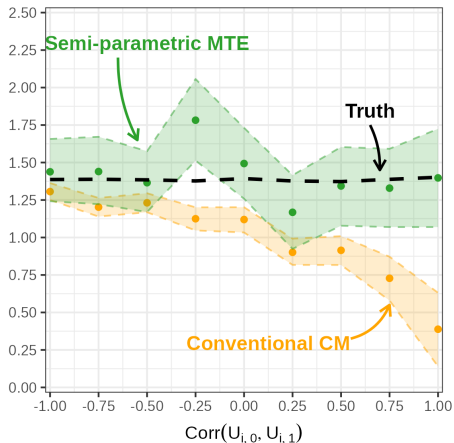
Appendix: CM with Selection — Estimation

Figure: CF Adjusted Estimates Work with Different Error Term Parameters.

(a) ADE.

(b) AIE.

Estimate



Estimate

